

GOVERNMENT OF BIHAR

Bihar State Disaster Management Authority

PROJECT PROPOSAL

Micro-Hydroelectric Power Generation Using Existing Road Culverts During Flood Season



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1 Executive Summary

The state of Bihar faces recurring challenges with seasonal flooding while simultaneously grappling with energy deficits that hinder economic development and quality of life for its citizens. This proposal presents an innovative and sustainable solution that addresses both challenges through the implementation of micro-hydroelectric systems integrated into existing road culverts across the state. Our proposed system leverages the natural flow of flood water through existing infrastructure to generate clean, renewable electricity without requiring additional land acquisition or major structural modifications.

The technology involves installing compact rotatory turbine systems within the culverts that are already constructed for flood water management, creating a dual-purpose infrastructure that serves both drainage and energy generation functions. The implementation of this technology represents a paradigm shift in how Bihar can approach flood management and energy generation simultaneously. Rather than viewing flood water as merely a challenge to be managed, this proposal transforms it into a valuable resource for sustainable energy production. The system is designed to be cost-effective, environmentally friendly, and scalable across the extensive network of road culverts throughout the state.

This initiative aligns perfectly with the Government of Bihar's vision for sustainable development and renewable energy adoption. The project promises to deliver significant economic benefits while contributing to environmental conservation and flood management efficiency. The technology is particularly well-suited for Bihar's geographical and climatic conditions, where seasonal flooding is a predictable phenomenon that can be harnessed for productive energy generation.

2 Introduction and Problem Statement

Bihar's geographical location in the Indo-Gangetic plain makes it particularly susceptible to seasonal flooding, with multiple rivers including the Ganges, Gandak, Kosi, and Bagmati flowing through the state. While these water bodies are crucial for agriculture and the region's ecosystem, they also pose significant challenges during monsoon seasons when flood waters need to be effectively managed through extensive drainage systems. The state's energy infrastructure faces parallel challenges with increasing demand for electricity driven by industrial growth, urbanization, and rural electrification initiatives. Traditional energy sources are becoming increasingly expensive and environmentally unsustainable, necessitating the exploration of alternative renewable energy solutions that can be implemented cost-effectively across the state.

Existing road infrastructure in Bihar includes thousands of culverts designed to facilitate the flow of flood water from one side of roads to the other. These culverts represent a significant investment in infrastructure that currently serves a single purpose of water management. However, the consistent and predictable flow of water through these structures during flood seasons presents an untapped opportunity for energy generation. The conventional approach to flood management treats water flow as a challenge to be contained and diverted, without considering its potential as an energy resource. Similarly, traditional approaches to renewable energy generation often require significant land acquisition and the construction of dedicated facilities, which can be cost-prohibitive and face resistance from local communities.

Our proposal addresses these interconnected challenges by proposing a solution that maximizes the utility of existing infrastructure while contributing to the state's renewable energy goals. The integration of electricity generation capabilities into existing culverts represents an innovative approach that can be implemented without disrupting existing flood management systems or requiring additional land resources.

3 System Diagram and Flowchart

3.1 System Diagram

The following diagram illustrates the proposed micro-hydroelectric power generation system integrated seamlessly into the existing road culvert infrastructure. This system is specifically designed to leverage the natural movement of flood water that passes through culverts during seasonal flooding. As depicted, the culvert continues to perform its essential role of flood water drainage, ensuring that agricultural lands and nearby settlements are protected from waterlogging and excessive flooding. However, in addition to its original purpose, the culvert now serves as a dual-purpose structure where strategically installed rotary turbine mechanisms capture the kinetic energy of the flowing water. These turbines are carefully positioned within the culvert to maximize energy conversion efficiency while allowing uninterrupted water flow. The system is designed to be adaptable to different culvert sizes and flow conditions typically found across Bihar, ensuring that the infrastructure remains fully functional for its drainage role even during peak flood events.

The diagram further demonstrates how the mechanical energy generated by the turbines is transferred through connected shafts and gear systems to high-efficiency electric generators positioned safely near the road. These generators convert the mechanical energy into usable electrical power, which is then conditioned and stabilized to meet grid standards. The produced electricity is transmitted via electrical cables to nearby existing transformers, facilitating the smooth integration of this new renewable power source into

Bihar's electricity distribution network. Importantly, the system is modular, meaning multiple units can be installed across a network of culverts to scale up power generation capacity as needed. By illustrating the system in this manner, the diagram provides a clear visual representation of how the proposed technology not only preserves the essential flood management capabilities of the culverts but also transforms them into valuable assets for clean, sustainable energy generation without the need for new land or major construction.



Figure 1: Micro-Hydroelectric Power Generation System Using Road Culverts Image generated using AI Tool (Gemini)

- **River Water Source:** The natural water flow from Bihar's river systems during flood seasons provides the primary energy source for electricity generation.
- **Road Culvert Infrastructure:** Existing culverts beneath roads serve as the installation site for the turbine systems, requiring minimal structural modifications.
- **Turbine Assembly:** Low-head turbines, such as Kaplan or cross-flow types,

are ideal for this setup as they operate efficiently with minimal height difference but high water flow. Installed directly inside the culvert, the turbine spins as water passes through, driving a connected shaft that powers a generator to produce **440V AC electricity**. This electricity is transmitted to an existing transformer for distribution to homes. The turbine assembly typically includes blades, a shaft, bearings, and a secure mounting frame. It must be durable, easy to maintain, and properly sized to handle varying flood conditions.

- **Electricity Generation Unit:** High-efficiency generators convert the mechanical energy from the turbines into electrical energy suitable for grid integration.
- **Power Grid Connection:** Generated electricity is conditioned and fed into the existing power grid infrastructure through standardized connection protocols.
- **Control and Monitoring Systems:** Smart monitoring systems provide real-time performance data and enable remote management of the installations.

The integrated design ensures that the primary function of flood water drainage is maintained while simultaneously generating renewable electricity. The system operates automatically during flood seasons when water flow is sufficient, and remains dormant during dry periods without impacting the culvert's drainage capacity.

3.2 Process Flowchart

The operational flow of the system is illustrated in the following flowchart, which demonstrates the step-by-step process from flood water intake to electricity generation and grid integration.

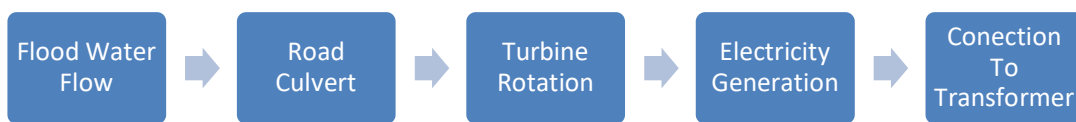


Figure 2: System Process Flow from Flood Water to Electricity Generation

4 Working Principle

The working principle of the proposed micro-hydroelectric system is based on the fundamental concept of converting the kinetic energy of flowing water into rotational mechanical energy, which is then transformed into electrical energy through electromagnetic induction. The system operates on the principle of impulse turbines, specifically designed for low-head applications where the water pressure is relatively modest but the flow volume is substantial. When flood water flows through the road culverts, it encounters the specially designed turbine blades positioned strategically

within the culvert channel. The flowing water strikes the curved turbine blades, imparting its kinetic energy to the rotor assembly. The blade design is optimized to maximize the momentum transfer from the water flow to the rotating shaft, ensuring efficient energy conversion even at varying flow rates typical of flood conditions. The rotational motion of the turbine is transmitted to a high-efficiency generator through a carefully engineered gear system that optimizes the rotational speed for maximum electrical output. The generator employs permanent magnet technology to ensure consistent power generation without the need for external excitation, making the system more reliable and maintenance-free in the challenging flood environment.

The electrical output from the generator is conditioned through power electronics that regulate voltage and frequency to match grid requirements. This conditioning system includes surge protection, harmonic filtering, and synchronization equipment that ensures the generated electricity meets utility standards for grid integration. The system also incorporates automatic disconnect features that protect both the generation equipment and the electrical grid during extreme weather conditions or maintenance operations. The entire system is designed to operate autonomously, with smart control systems that monitor water flow conditions, electrical output, and equipment health. During periods of insufficient water flow, the system automatically enters a standby mode, consuming minimal power while remaining ready to resume generation when flow conditions improve. This intelligent operation ensures maximum efficiency and equipment longevity while minimizing operational complexity and maintenance requirements.

5 Proposed Technology and Methodology

The proposed system involves the installation of compact, efficient rotary turbine mechanisms within existing road culverts to harness the kinetic energy of flowing flood water. The technology is specifically designed to operate effectively with the variable flow rates and water levels characteristic of seasonal flooding in Bihar. The core component of the system is a low-head turbine specifically engineered for installation in culvert environments. These turbines are designed to operate efficiently with relatively low water pressure and variable flow rates, making them ideal for the seasonal and unpredictable nature of flood water flow. The turbine design incorporates advanced materials and engineering principles to ensure durability and minimal maintenance requirements in the challenging environment of flood-prone areas. The installation process involves minimal modification to existing culvert structures, preserving their primary function of flood water management while adding electricity generation capabilities. The turbine systems are designed to be modular and scalable, allowing for customization based on the specific dimensions and flow characteristics of individual culverts.

The electrical generation system includes advanced power conditioning equipment to ensure consistent electricity output despite variable water flow conditions. The system incorporates battery storage capabilities to provide continuous power availability even during periods of low or no water flow, ensuring reliability for connected communities and facilities. Integration with existing electrical grid infrastructure is facilitated through standardized connection protocols and safety systems that ensure seamless integration with Bihar's electrical distribution network. The system is designed to meet all relevant electrical safety standards and grid connection requirements, ensuring compatibility with existing infrastructure and regulatory frameworks. The technology also includes comprehensive monitoring and control systems that provide real-time data on electricity generation, water flow rates, and system performance. This data collection capability supports both operational optimization and long-term performance analysis, enabling continuous improvement of the system's effectiveness and efficiency.

6 Technical Specifications and Design Considerations

The proposed turbine systems are specifically designed for Bihar's culvert environments, where low-pressure but high-flow water conditions are common during seasonal flooding. These turbines use advanced, low-head optimized blade designs that ensure maximum energy extraction even under variable flow rates. The blade geometry and materials are carefully selected to minimize energy loss, resist corrosion, and handle debris typically found in flood waters, ensuring durability and low maintenance. The compact design allows the turbines to fit within existing culverts without disrupting their primary drainage function, making them ideal for widespread deployment across Bihar's road network

S.no.	Component	Specification	Approximate Cost (INR)
1	Water flow passage (culvert)	4 to 4.5 feet diameter (already installed)	Already installed (Assuming)
2	Water turbine (rotary part)	Low-head, suitable for culvert flow; capacity: 5-10 kW	90,000
3	Generator	440V AC, 3-phase, synchronous/induction, 5-10 kW	1,25,000
4	Power transformer	Step-up/down, 440V AC, 50 Hz, 3-phase	Already installed (Assuming)
5	Control & protection system	Voltage regulator, circuit breakers, protection relays	50,000
6	Power transmission & wiring	Cables, connectors, switchgear for grid connection	40,000
7	Installation & deployment	Civil works, electrical installation, commissioning (Bihar)	50,000
Total Estimated Cost (excluding culvert):			3,55,000

The modular design of the system allows for easy installation, maintenance, and potential expansion of capacity at individual sites. This modularity also facilitates standardization of components across multiple installations, reducing manufacturing costs and simplifying maintenance procedures.

7 Benefits and Impact Analysis

The implementation of this technology offers numerous benefits that extend beyond simple electricity generation. The primary benefit is the conversion of existing flood management infrastructure into dual-purpose systems that continue to serve their original drainage function while simultaneously generating renewable energy. Environmental benefits include the reduction of carbon emissions associated with traditional electricity generation methods, contributing to Bihar's environmental sustainability goals. The technology produces no harmful emissions during operation and has minimal environmental impact compared to traditional power generation facilities. Economic benefits encompass both direct and indirect impacts on the state's economy. Direct benefits include the generation of electricity that can be sold to the grid or used to power local facilities, creating a new revenue stream for the state. Indirect benefits include the potential for job creation in manufacturing, installation, and maintenance of the systems.

Social benefits include improved access to electricity in rural and flood-prone areas that may have limited grid connectivity. The distributed nature of the generation system can provide a resilient power supply to communities that might otherwise experience frequent power outages during flood seasons. The technology also contributes to flood management effectiveness by maintaining and potentially improving the flow characteristics of existing culverts. The turbine systems can be designed to actually improve water flow dynamics while generating electricity, creating a synergistic relationship between energy generation and flood management. Long-term benefits include the development of local expertise in renewable energy technology, potentially positioning Bihar as a leader in innovative flood water utilization techniques. This leadership position could attract additional investment and research opportunities in related technologies.

8 Implementation Strategy and Timeline

The implementation of this technology across Bihar's extensive culvert network requires a phased approach that allows for testing, optimization, and scaling of the system. The initial phase involves the selection of pilot sites that represent different types of culverts and flow conditions across the state. Pilot site selection will be based on a comprehensive analysis of existing culvert infrastructure, including flow rates, structural

conditions, and accessibility for installation and maintenance. The pilot phase will include detailed monitoring and performance analysis to optimize the technology for Bihar's specific conditions. The second phase involves the expansion of the technology to additional sites based on the success and lessons learned from the pilot installations. This phase will include the development of standardized installation procedures and the establishment of local manufacturing and maintenance capabilities.

The final phase encompasses the state-wide rollout of the technology across all suitable culvert locations. This phase will be supported by comprehensive training programs for local personnel and the establishment of ongoing maintenance and support systems. Each phase includes specific milestones and performance targets that will be used to evaluate the success of the implementation and guide decisions about subsequent phases. The timeline is designed to be flexible and responsive to the lessons learned during implementation while maintaining steady progress toward the overall objectives. Community engagement and stakeholder involvement are integral components of the implementation strategy. Local communities will be involved in the planning and implementation process to ensure that the technology meets their needs and has their support.

9 Economic Analysis and Funding Requirements

The economic viability of this project depends on several factors including installation costs, electricity generation capacity, maintenance requirements, and the economic value of the generated electricity. The distributed nature of the installation sites provides opportunities for cost optimization through economies of scale in manufacturing and installation. Initial capital investment requirements include the cost of turbine systems, electrical components, installation labour, and supporting infrastructure. These costs are offset by the utilization of existing culvert infrastructure, eliminating the need for new civil construction that would be required for traditional hydroelectric installations. Operational costs include regular maintenance, monitoring system operations, and periodic component replacement. The robust design of the system minimizes maintenance requirements while maximizing operational lifespan, contributing to favourable long-term economics. Revenue generation potential includes direct electricity sales to the grid, reduced electricity costs for state facilities, and potential carbon credit opportunities associated with renewable energy generation. The distributed nature of the generation also provides value through improved grid resilience and reduced transmission losses.

Funding options include state government investment, central government renewable energy programs, international development funding, and potential private sector

partnerships. The project's alignment with multiple government priorities including renewable energy, flood management, and rural development makes it eligible for various funding sources. Return on investment calculations demonstrate favourable economics over the system's operational lifetime, with payback periods that compare favourably to other renewable energy technologies. The project's benefits extend beyond direct financial returns to include social and environmental value that enhances the overall economic case.

10 Environmental and Social Impact Assessment

The environmental impact of this technology is overwhelmingly positive, with the primary impact being the generation of clean, renewable electricity that displaces fossil fuel-based power generation. The technology produces no direct emissions during operation and has minimal impact on the existing drainage infrastructure and surrounding environment. The installation process is designed to minimize environmental disruption, with most work conducted during dry seasons when culverts are not actively draining flood water. The systems are designed to be removable if necessary, ensuring that they do not permanently alter the existing infrastructure. Social impacts include improved access to electricity in rural and underserved communities, potential job creation in manufacturing and maintenance, and the development of local technical expertise in renewable energy systems. The technology also contributes to community resilience by providing distributed power generation that can continue operating during grid outages.

The project includes comprehensive consultation with local communities to ensure that the technology meets their needs and addresses any concerns about the implementation. Community feedback will be incorporated into the design and implementation process to maximize social benefits and minimize any negative impacts. Environmental monitoring programs will be established to track the long-term environmental impact of the installations and ensure that they continue to operate within acceptable environmental parameters. These monitoring programs will also provide data for optimizing the technology and informing future implementations.

11 Risk Assessment and Mitigation Strategies

The implementation of this technology involves several categories of risk that must be carefully managed to ensure project success. Technical risks include the possibility of equipment failure, performance below expectations, or compatibility issues with existing infrastructure. Technical risk mitigation strategies include comprehensive testing during the pilot phase, the use of proven technologies and materials, and the establishment of

robust maintenance and support systems. Redundancy is built into critical system components to ensure continued operation even if individual components fail. Environmental risks include potential impacts on water flow patterns, the possibility of equipment damage during extreme weather events, and the potential for environmental contamination if systems are damaged. These risks are mitigated through careful system design, regular monitoring, and comprehensive safety systems.

Financial risks include cost overruns, lower than expected performance, and changes in electricity market conditions that could affect project economics. These risks are managed through careful project planning, comprehensive cost estimation, and the development of flexible implementation strategies that can adapt to changing conditions. Political and regulatory risks include changes in government policy, regulatory requirements, or public support for the project. These risks are mitigated through comprehensive stakeholder engagement, compliance with all applicable regulations, and the development of broad-based support for the project. Operational risks include maintenance challenges, personnel safety concerns, and the possibility of vandalism or theft of equipment. These risks are addressed through comprehensive training programs, safety protocols, and security measures appropriate for the rural and distributed nature of the installations.

12 Conclusion and Recommendations

This proposal presents a unique opportunity for the Government of Bihar to address multiple challenges simultaneously through the implementation of innovative flood water electricity generation technology. The integration of electricity generation capabilities into existing culvert infrastructure represents a cost-effective and environmentally sustainable approach to renewable energy development. The technology offers significant benefits including renewable electricity generation, improved flood management, economic development opportunities, and enhanced community resilience. The distributed nature of the generation system provides additional benefits in terms of grid stability and reduced transmission losses. The implementation strategy provides a structured approach to technology deployment that minimizes risk while maximizing the potential for success. The phased implementation approach allows for learning and optimization while building toward state-wide deployment of the technology.

We recommend that the Government of Bihar approve this proposal and provide the necessary support for initial pilot implementations. The pilot phase will provide valuable data and experience that will inform the broader implementation strategy and demonstrate the technology's potential for addressing Bihar's energy and flood management challenges. The success of this project has the potential to position Bihar as a leader in innovative renewable energy technology while providing significant benefits

to the state's citizens and economy. We are prepared to work closely with the government to ensure the successful implementation of this technology and the realization of its full potential. We request the opportunity to present this proposal in detail to the relevant government departments and to begin the planning process for pilot implementations. The technology represents a significant opportunity for Bihar to demonstrate leadership in sustainable development while addressing critical infrastructure and energy challenges.

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